

POPSCRIPT LANGUAGE SPECIFICATION

Version 1.0.4 | Official Technical Reference

1. Core Syntax & Comments

PopScript is an interpreted language with explicit block delimiters. All instructions are processed top-to-bottom.

Comments: Single-line comments start with `$/`.

```
$/ This is a standard comment
```

2. Data Types

Type	Keyword	Description
Integer	<code>int</code>	64-bit signed integers.
Float	<code>float</code>	64-bit floating point numbers.
String	<code>string</code>	UTF-8 character sequences.
Boolean	<code>bool</code>	Values: <code>true</code> or <code>false</code> .
List	<code>list</code>	Ordered dynamic collection.

3. Control Structures

Conditionals

Blocks are terminated using the `stop;` keyword.

```
if x > 10;
  print("Greater than 10")
stop;
```

Loops

While Loop (when): Continues execution as long as the condition is true.

```
when status == true;
    $/ execution logic
stop;
```

Iteration (for): Used for ranges and list traversal.

```
for i with num(10); $/ Executes 10 times
    print(i)
stop;

for item with myList; $/ Iterates over elements
    print(item)
stop;
```

4. Functions

Functions are declared with `func`. Use `export` for library visibility. Returns are handled via `return`.

```
func calculate(a, b);
    return a + b
stop;

export func process_data(data);
    $/ logic
stop;
```

5. Memory Management (ARMemory)

The `armemory` directive (default: false) enables manual memory control. If used, it must be the first instruction.

```
armemory true
alloc(ptr, 512)
free(ptr)
```

6. Built-in Modules

- **ui:** Windowing, labels, buttons, and inputs.

- **random:** `random.number(from, to)`.
- **file:** File I/O operations.
- **time:** System time functions.

```
lib import ui
ui.window(title="App", width=300, height=200)
ui.run()
```